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Special Issue: Comparative Approaches to Studying Privacy: Opening Up New Perspectives

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AI Privacy in Context: A Comparative Study of Public and Institutional Discourse on Conversational AI Privacy in the US and Chinese Social Media

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Abstract

The proliferation of conversational artificial intelligence (AI) systems, such as chatbots, has sparked widespread privacy concerns. Previous research suggests that privacy perceptions and practices vary across sociocultural contexts. This study examines public and institutional discourses on conversational AI privacy in the United States and China. Semantic network and discourse analyses of privacy-related discussions on Twitter and Weibo reveal divergent patterns. On Twitter, public discourse emphasizes privacy risks and concerns and advocates for systemic changes, while institutional discourse promotes individualistic approaches to privacy protection. Conversely, on Weibo, public discourse is less focused on privacy risks and more on the positive impacts of AI, aligning closely with institutional narratives. These variations are intertwined with the cultural, political, economic, and regulatory contexts of the two countries. Our study underscores the importance of multi-level analysis in comparative privacy research to provide a holistic view of privacy in various contexts.

Keywords

comparative privacy, AI ethics, generative AI, semantic network analysis, discourse analysis

Introduction

Conversational artificial intelligence (AI) systems (e.g., smart speakers, chatbots) have permeated our daily lives, providing services from customer service to entertainment to mental health support (Li et al., 2023; Liao et al., 2018; Xu et al., 2017). Recently, the rise of large language models (LLMs) such as ChatGPT has enabled AI chatbots to interact more naturally with users, potentially leading to exponential disclosure of personal and intimate information, both voluntarily and unknowingly (Laestadius et al., 2022; Lee et al., 2020; Zhang et al., 2024). This has raised a host of ethical concerns, notably around privacy and surveillance (Kretzschmar et al., 2019; Sağlam et al., 2021). Existing studies about chatbot privacy have primarily focused on user perspectives, emphasizing how chatbot features and user characteristics influence privacy concerns and behaviors (Ischen et al., 2020; Sağlam et al., 2021).

While the pursuit of privacy is universal (Altman, 1975), its norms, perceptions, and practices vary across sociocultural contexts (Moore, 1984; Zarrow, 2002). For instance, privacy in China is often intertwined with social roles and personal relationships (Yuan et al., 2013), whereas in Qatar, it is viewed as a moral value that extends into the afterlife (Abokhodair et al., 2016). Extensive research has uncovered notable cultural differences in privacy attitudes and practices (Cho et al., 2023; Krasnova et al., 2012; Trepte et al., 2017), with higher privacy concerns in individualistic cultures, such as the United States and Germany, than in more collectivist societies, such as China and Singapore (Cho et al., 2009; Wang et al., 2011).

While insightful, existing comparative privacy research (CPR) often views country as the unit of analysis and empathizes cultural values as explanatory factors, which

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may risk simplifying cultural complexities and obscuring the exploration of other important sources of variation (Baskerville, 2003; Livingstone, 2013; McSweeney, 2002). In fact, privacy perceptions and practices are shaped by a multitude of factors, including regulatory frameworks, political and economic systems, and cultural norms (Bennett, 2013; Vitak et al., 2022; Yuan et al., 2013). The CPR framework (Masur et al., 2021) emphasizes the need to move beyond a sole focus on country-specific comparisons and to identify meaningful units of comparison while also examining the interplay between various contextual factors in shaping privacy (Masur et al., 2021).

Examining discourse is crucial for understanding how privacy is valued, perceived, and enacted in specific contexts (Fairclough, 2003; Matassi & Boczkowski, 2021). While much CPR has focused on public perceptions of privacy (Abokhodair et al., 2016; Liang et al., 2017; Yuan et al., 2013), less is known about the discourses of institutions, such as corporations, media, and nongovernmental organizations (NGOs), which play a crucial role in setting social agendas and shaping public opinions (Fu & Zhang, 2019; McCombs et al., 2014; Vargo et al., 2018). Investigating institutional discourse can reveal how these entities enact and communicate privacy-related practices and policies, elucidating the meso-level contexts of conversational AI privacy. Moreover, the interplay between public and institutional discourse may influence policy-making and societal norms (Boykoff & Roberts, 2007; Wakefield et al., 2010). Thus, examining both public and institutional discourses can provide a comprehensive understanding of privacy and help distinguish factors producing variations in privacy between nations (McLeod & Lee, 2013).

This study examines public and institutional discourse on conversational AI privacy on social media platforms in the United States and China. These two countries were chosen due to their leading positions in the global AI landscape and distinct sociocultural, political, and economic contexts. Using semantic network analysis (SNA) and discourse analysis, this research explores how privacy issues regarding conversational AI are discussed by individuals and institutions on Twitter and Weibo, and the interplay between these discourses. Guided by the CPR framework (Masur et al., 2021), we interpret findings within the cultural, political, economic, and legal contexts of countries. This multi-level analysis provides a nuanced understanding of AI privacy, offering insights into variations in privacy concerns and expectations, potentially informing culturally and contextually appropriate privacy policies and practices.

Literature Review

Public Perceptions of AI Risks

The penetration of AI into various domains, such as healthcare, education, and finance, has brought about noticeable changes in society (Chen et al., 2020; Darcy et al., 2016; Frank et al., 2019). Meanwhile, it has engendered various ethical concerns and risks (CAIS, 2023; Crawford, 2021; Zuboff, 2019). Key concerns include fears of diminished privacy, resulting from data breaches, misuse of personal information, unregulated system surveillance, and opaque algorithmic operations (Kelley et al., 2021). Compounded by a perceived loss of personal data control, these concerns can deter users from using intelligent systems (Prakash & Das, 2020). Examining public opinions on AI can shed light on factors influencing user acceptance and inform the responsible development of AI systems that align with societal values and address user concerns (Cave et al., 2019).

Existing research reveals notable differences in attitudes toward AI across countries and regions. For instance, a recent study examines public perceptions of AI conversational agents in the United States and China, suggesting that Chinese users tend to view AI agents more positively, empathizing their hedonic and socio-emotional utility, whereas U.S. users held an ambivalent attitude, focusing more on the functional utility of AI agents (Liu et al., 2024). Moreover, a multi-country survey found that developed countries generally expressed worries about AI, while developing nations found AI exciting and useful (Kelley et al., 2021). A subsequent survey involving 17 countries showed an overall hesitance to trust AI systems, with the lowest acceptance levels in countries such as Finland and Japan, and the highest in emerging economies such as Brazil, India, China, and South Africa (Gillespie et al., 2023). The survey also reported globally consistent concerns about AI, among which cybersecurity and privacy breaches were the most pressing. These findings highlight the need for a context-specific understanding of AI perceptions, particularly regarding privacy risks.

Conversational AI systems such as voice assistants and chatbots, due to their conversational nature and widespread use, often amass extensive user data and prompt users to disclose personal information (Lee et al., 2020), heightening privacy concerns (Ischen et al., 2020; Jo et al., 2024). A recent study revealed that LLM-based conversational agents generate significant privacy concerns among users, yet the design features of such technologies often impede users' ability to comprehend and manage privacy risks (Zhang et al., 2024). Despite these findings, there is limited knowledge about how privacy concerns related to conversational AI vary across cultures, which hampers a nuanced understanding of user preferences and expectations regarding AI privacy. Therefore, this study aims to explore perceptions of privacy issues related to conversational AI across countries. In the following section, we provide an overview of the literature on privacy from a comparative perspective.

Comparative Privacy Research

Privacy has received considerable scholarly attention, yet its definition is often nebulous and inconsistent. Privacy has been diversely construed as a legal right, a form of

control, a state of seclusion, and a dynamic process of boundary management (Altman, 1975; Trepte & Masur, 2023; Westin, 1967). Much privacy research is rooted in Western philosophies of liberty, emphasizing freedom from external influences (Locke, 1689; Mill, 1859), framing privacy as a cornerstone for individual autonomy and self-realization (Capurro, 2005; Whitman, 2004). However, this liberal, individualistic perspective might not fully capture privacy conception and practices in diverse sociocultural contexts. Previous research indicates that the definition and value of privacy are deeply intertwined with sociocultural factors, such as cultural norms and religious beliefs. For instance, Yuan et al. (2013) found that privacy discourse in Chinese language often revolved around social roles and personal relationships, despite individualistic expressions about privacy. Abokhodair et al. (2016) explored privacy in an Arabic-language context and found that the notion of privacy was predominantly communal and often seen as a moral value, deeply influenced by religious, gender, and social factors in Qatar.

Given the context-dependent nature of privacy (Nissenbaum, 2011), a plethora of research has examined cross-country and cross-cultural differences in privacy perceptions and practices (e.g., Cho et al., 2009, 2023; Trepte & Masur, 2016; Wang et al., 2011). The results provided valuable but mixed insights. For instance, some studies found that people from individualistic cultures, such as the United States, had greater concerns about online privacy than those from collectivistic cultures, such as China and India (Cho et al., 2009; Kumaraguru & Cranor, 2006; Wang et al., 2011). However, other studies have found no significant correlation or even contradictory results (Cho et al., 2023; Liang et al., 2017; Trepte et al., 2017). The varied findings can be partially attributed to the limitations of using Hofstede's (1984) cultural dimensions, such as individualism and uncertainty avoidance, as frameworks to explain variations in privacy perceptions across countries. Hofstede's model has been critiqued for equating nation with culture, treating cultures as monolithic, and overlooking the dynamic and fluid nature of cultural identities and practices (Baskerville, 2003; Livingstone, 2013; McSweeney, 2002). In addition, it is argued that besides national culture, other factors such as regional, organizational, or professional cultures also play a significant role in shaping individual values and behaviors (McSweeney, 2002).

Recent research has highlighted the impact of other macro-level factors, such as regulatory frameworks, on privacy perceptions (Vitak et al., 2022). Bellman et al. (2004) found that governmental regulation directly impacts information privacy concerns across countries. For example, the enactment of the European Union's General Data Protection Regulation (GDPR) has been shown to lower privacy concerns on average (Paul et al., 2020). This might explain why Dutch participants, despite being from an individualistic culture, reported lower privacy concerns than Americans (Vitak et al., 2022). Interestingly, before GDPR, German social media users demonstrated more cautious attitudes toward privacy risks and engaged in more privacy-protective behaviors than their American counterparts (Trepte & Masur, 2016). These findings underline the complexity of privacy perceptions and the need for a more holistic framework to better understand and interpret them.

The CPR framework (Masur et al., 2021) was proposed to address these challenges. The CPR framework foregrounds the embeddedness of privacy perceptions and practices and critiques existing research for overlooking contextual factors at different levels in explaining varied privacy perceptions. This framework provides a systematic approach to assessing the comparability of the object under study, identifying meaningful units of comparison, and scrutinizing the interplay between these units of comparison and privacy. The authors suggested five categories of factors—cultural, social, political, economic, and technological—as fruitful units of comparison to explain variability in privacy perceptions, attitudes, and practices. These contextual factors may directly and interactively shape various aspects of privacy across multiple levels (Masur et al., 2021).

AI Privacy in the United States and China

This study examines and compares perceptions of conversational AI privacy in the United States and China. Following the CPR framework, this study first evaluates the comparability of the object of investigation—privacy perceptions regarding conversational AI—in these two countries. Given that language can construct, maintain, and challenge social realities (Fairclough, 2003), we delve into discourse on conversational AI privacy to understand how it is perceived, interpreted, and enacted in these two contexts. While previous studies showed that privacy may carry varied cultural meanings and values (Abokhodair et al., 2016; Yuan et al., 2013), we believe that discourse on domain-specific privacy in the US and China is functionally equivalent, allowing for meaningful comparison.

Both countries are on par in terms of conversational AI development. They both have initiated strategic national AI programs to drive economic growth and address societal challenges. In 2019, the White House unveiled the American Artificial Intelligence Initiative, establishing a national strategy to sustain American leadership in AI (Parker, 2020). Similarly, China launched the New Generation of Artificial Intelligence (NGAI) Development Plan (2015–2030) in 2017, marking its first national-level effort to systematically promote AI innovation and establish AI governance standards, with the goal of becoming a global AI leader (Wu et al., 2020; Roberts et al., 2021). Moreover, conversational AI has seen significant interest and market growth in both countries, making the US and China the world's leading users of smart speakers (McNair, 2019). While the US had a head start on smart speaker adoption with the launch of Amazon Echo in 2014, China has ascended to the forefront

Table 1. Units of Comparison Between the United States and China Regarding Conversational AI Privacy.

Category	Definition	U.S.	China
Cultural	Shared beliefs, values, norms, and practices	Individualism, valuing individual rights and autonomy	Collectivism, valuing collective welfare and harmony
Political	Structures, processes, and activities by which a country is governed	Democracy; low state surveillance	Authoritarianism; high state surveillance
Economic	Economic systems; market structures; infrastructure & investment	Capitalist, free-market economy; strong IT infrastructure; heavy investment in AI	Socialist market economy; strong IT infrastructure; heavy investment in AI
Technological	Technology & infrastructure	Strong infrastructure; heavy investment in AI	Strong infrastructure; heavy investment in AI; AI as a national priority
Legal/Regulatory	Laws, regulations, and legal frameworks	State and sector-specific privacy laws	Government privacy laws

of the smart speaker market, surpassing the US in leading the smart speaker market (McNair, 2019; QuestMobile, 2023). This substantial growth is fueled by the substantial investments from major Chinese IT corporations in conversational AI, leading to tremendous advancements in chatbots and voice assistant technologies (Fan, 2023).

Despite similarities, the US and China differ vastly in cultural, political, legal, and economic contexts, which are likely to shape privacy issues related to conversational AI. Culturally, the US is known for its individualism, emphasizing personal autonomy and individual rights, as opposed to the collectivist-oriented Chinese culture that cherishes collective welfare, familial respect, and harmony. These cultural differences may lead to varying views on privacy. Politically, the democratic governance of the United States contrasts starkly with China's socialist one-party state, resulting in different privacy and AI policies (Hine & Floridi, 2022). The US has state- and sector-specific privacy laws, such as the California Consumer Privacy Act (CCPA) and Health Insurance Portability and Accountability Act (HIPAA) but lacks a federal data privacy legislation. China's centralized approach includes the Personal Information Protection Law (PIPL), which, despite its comprehensiveness, allows exceptions for state security and public interests, facilitating state surveillance. Economically, the US's free-market economy means that private enterprises drive technological innovation for competitive advantage and profitability. In contrast, China's mixed socialist-market economy compels tech companies to work in tandem with the government's objectives. Table 1 illustrates these units of comparison.

Public vs. Institutional Discourse on AI Privacy

The discourse on AI privacy is constituted by various stakeholders, including users, media, corporations, and governments. Each group offers unique perspectives and attitudes, creating a rich and complex narrative of privacy (Du & Xie, 2021; McStay, 2020). Prior work has shown that different social actors often use different frames to promote particular attitudes or interpretations (Hon, 2016; Pan & Kosicki, 2001; Vliegenthart & Van Zoonen, 2011). For instance, climate change activists and mainstream media adopt different framing strategies, selectively sharing content congruent with their beliefs (Chen et al., 2023). Despite much comparative research on public perceptions of privacy (Abokhodair et al., 2016; H. Liang et al., 2017; Ma, 2023), institutional perspectives have received less scrutiny, resulting in a partial understanding of the privacy landscape and the interplay among various stakeholders.

Public discourse and institutional discourse are two distinct forms of communication, each with its own characteristics and functions within society. Public discourse refers to the open exchange of ideas, opinions, and information among the general public (Habermas, 1989). It takes place in public spaces and forums, such as social media, where individuals can freely discuss and debate various issues. Public discourse is characterized by inclusivity and diversity, allowing participation from a wide range of individuals and groups, and it can be either informal (e.g., social media conversations) or formal (e.g., public debates, town hall meetings). It plays a critical role in democratic societies by enabling citizens to express their views, influence public opinion, and participate in the decision-making process (Habermas, 1989; Li et al., 2021).

Institutional discourse, in contrast, occurs within or is produced by institutions, such as governments, corporations, educational institutions, and other formal organizations. Institutional theory posits that institutions provide stability and meaning to social life through regulative, normative, and cultural-cognitive elements (Hodgson, 2006; Scott, 2008). Prior work shows that institutions, such as NGOs and media, play a vital role in setting social agenda and shaping public opinions (Fu & Zhang, 2019; McCombs et al., 2014; Vargo et al., 2018). Institutional discourse is typically structured and governed by specific rules and protocols, often controlled by individuals or groups in positions of authority within the institution. The communication is typically aimed at achieving specific organizational objectives, such as policy implementation, corporate strategy, or educational goals (Fu & Zhang, 2019).

McLeod and Lee (2013) contend that comparative communication research typically focuses either on individuals or on media and political institutions, but seldom integrates both. This disconnection risks producing explanatory models that are incomplete and potentially misleading. Examining both public and institutional discourses allows for a comprehensive understanding of privacy concerns, experiences, and practices. While public discourse highlights what individuals and communities worry about, institutional discourse shows how these concerns are addressed at policy and organizational levels.

Privacy-related discussions on social media platforms, specifically Twitter and Weibo, offer a fertile ground for studying discourse on conversational AI privacy. This study focuses on these platforms due to their notable comparability. First, their user base sizes are closely aligned, with Twitter boasting 238 million and Weibo 252 million daily active users (Statista, 2022; Weibo, 2022). These two platforms also feature a similar user demographics, with half of users being under the age of 35, and more than two-thirds of the users being male (Similar web, 2023; Duarte, 2023). Second, both are microblogging services with similar technical features and functionalities. They utilize a threaded format for posts and comments, support multimodal media content, and include interactive elements such as mentions, likes, reposts. Hashtags, a key feature on both platforms, help categorize topics, enhancing content visibility and engagement with trending topics. Moreover, both Twitter and Weibo serve as platforms for diverse social actors to voice their opinions on a variety of public issues. These platforms have been extensively employed to investigate public discourses and debates on contentious issues, such as climate change (Chen et al., 2023), COVID-19 vaccines (Luo et al., 2021), gene editing (W. Wang & Guo, 2018), Black Lives Matter movement (Keib et al., 2018), and the #MeToo movement (Xiong et al., 2019).

To explore the variations in AI privacy discourse from individuals and institutions in the US and in China, we ask:

> **Research Question 1 (RQ1):** How does public discourse on conversational AI privacy vary across the US and Chinese social media platforms?

> **Research Question 2 (RQ2):** How does institutional discourse on conversational AI privacy vary across the US and Chinese social media platforms?

Further investigation into the interplay of institutional discourse and public perceptions of AI privacy could also bear fruit. A cross-level analysis can shed light on the dynamics between forces at a different level of abstraction, which helps "distinguish the social origins and processes that are working to produce similarities and differences between nations" (McLeod & Lee, 2013, p. 434). Public and institutional discourses are often interdependent. On one hand, public discussions about AI privacy issues may drive policy changes and organizational practices. For instance, the implementation of the GDPR in Europe was significantly influenced by public concerns about privacy and data protection. On the other hand, institutions can shape individuals' behaviors, attitudes, or beliefs (Meyer & Rowan, 1977; Scott, 2008). For instance, cooperate policies and practices induce digital resignation—a sense of futility toward privacy protection (Draper & Turow, 2019). This often results in a privacy paradox, where individuals continue using digital services despite privacy concerns due to perceived benefits (Barnes, 2006; Dienlin & Trepte, 2015).

The interplay between public and institutional discourse can vary across sociopolitical contexts. For instance, democratic systems may foster a dynamic relationship between the public and institutions, with public discourse influencing institutional policies and practices, which in turn affect public perceptions and behaviors (Bennett & Raab, 2006). In authoritarian regimes, however, institutional discourse is more likely to dominate and shape public perceptions, whereas public discourse hardly influences institutional policies (King et al., 2013). Understanding these dynamics is crucial for developing effective privacy policies and practices that are culturally and contextually appropriate. Examining the convergence and divergence of public and institutional discourse on privacy can deepen our understanding of the complex dynamics influencing privacy. It also helps identify gaps or mismatches between public expectations and institutional practices, which is crucial for effective governance and corporate responsibility. Thus, we ask:

> **Research Question 3 (RQ3):** How do public and institutional discourses on conversational AI privacy converge and diverge on the US and Chinese social media platforms?

Methods

To answer our research questions, we collected publicly available tweets/posts from Twitter and Weibo. Twitter's API access was deployed for tweets retrieval. As Weibo restricts its API access to independent researchers, we resorted to the platform's advanced keyword-based search function to obtain pertinent posts, in line with the methodologies of previous studies (Chen et al., 2019; Li et al., 2021; Luo et al., 2021).

Data collection was conducted via several stages. First, a preliminary list of keywords was summarized and generated using an iterative term discovery method based on the initial seed terms: "chatbot" and "聊天机器人" ("chatbot" in Chinese). This process involved two researchers performing open coding on 1,000 chatbot-related posts randomly selected from Twitter and Weibo corpus, respectively, identifying frequently used terms closely related to "chatbot" in both English

and Chinese. Through several rounds of iterative open coding and discussions, an initial list of keywords was identified until no further relevant keywords emerged. Subsequently, we expanded our keyword catalog by including the names of top 10 chatbot apps in the US and China, respectively, based on the download metrics provided by Qimai (qimai.cn), a mobile app database housing app download data from multiple countries (Fan et al., 2022; Wang et al., 2022). This resulted in additional 35 and 47 keywords in English and Chinese, respectively. Exemplar keywords include "voice assistants," "conversational agent," "AI companion," and "chatgpt." The full list can be found in Supplemental Appendix 1.

Next, posts containing any of those keywords on Twitter and Weibo between July 20, 2017, and January 30, 2023, were retrieved. We selected the start date because it was when China launched its "New Generation Artificial Intelligence Development Plan," a blueprint for a complete AI ecosystem for the country (Wu et al., 2020), which presumably has sparked public discussions of AI. In addition, 2017 also marked the surge of the number of chatbots and market growth in chatbot usage was observed in the United States (Adamopoulou & Moussiades, 2020). The end date was set as the day preceding our final data scraping. To determine relevant tweets from the United States, we relied on both GPS location and user profile data to validate the residence location (for further details, please see Supplemental Appendix 2).

Through the above-mentioned process, 755,399 tweets and 229,990 Weibo posts were retained. Several rounds of duplicate removal and noisy data preprocessing were conducted (details can be found in Supplemental Appendix 3), retaining 754,144 tweets and 200,049 Weibo posts. In addition, we removed posts published by AI applications' official social media accounts (e.g., "ReplikaAI" on Twitter) to eliminate advertisements and posts obviously generated by AI bots. After this, our datasets consisted of 753,807 tweets and 199,089 Weibo posts (see Table B in Supplemental Appendix 3).

To identify privacy-related posts, we utilized a conservative keyword strategy, focusing on "privacy" and its Chinese equivalent "隐私," as suggested by prior research (Yuan et al., 2013; Ma, 2023). This resulted in a dataset of 1,165 tweets and 1,134 Weibo posts for further analysis. Our keyword search method was selected based on a careful evaluation of the balance between data relevance and comprehensiveness. While using specific keywords might overlook some pertinent data, broader terms often introduce noise and irrelevant content. To validate our approach, we undertook supplementary data collection using a wider range of privacy-related terms (e.g., surveillance, data, misuse), which showed less than 10% of these posts were relevant. Further details about this validation process are available in Supplemental Appendix 4. The final corpus of tweets and Weibo posts were reviewed carefully by two researchers from the team to ensure their relevance to the study.

To differentiate between public and institutional discourse, we categorized users on Weibo and Twitter into two types: individuals and institutions (i.e., media, corporations, schools, NGOs, and government agencies). Given the distinct classification systems of these platforms, we used different approaches for each. On Weibo, we utilized the platform's pre-existing user categorization, grouping them into individual users (ordinary users, influencers, and micro-influencers) and institutional users (media, corporate, school, NGO, and government). For Twitter, which lacks a detailed classification system, we manually examined the profiles of 1,001 users to categorize them as individual or institutional (media, corporate, school, NGO, and government). Details of these classification methods are available in Supplemental Appendix 5.

Semantic Network Analysis

SNA was applied to elucidate the primary themes and concepts in privacy discourses on Twitter and Weibo. SNA is a computer-assisted content analysis method that examines the associations among words (or co-occurrence) to identify patterns, prevalent themes, and underlying meanings within the text (Segev, 2021). The basic assumption behind this approach is that the words we use reflect our cognitive processes (Woelfel & Fink, 1980) and "unveil hidden meanings and desires" (Segev, 2021, p. 2). For instance, Yuan et al. (2013) applied SNA to the discourse of privacy on Weibo to understand the meanings and values related to privacy in contemporary China. Luo et al. (2021) employed SNA to examine public perspectives of COVID-19 vaccine on U.S. and Chinese social media. Following the methods in these studies, this study performed SNA through multiple steps, including data preprocessing, semantic metrics generation, cluster analysis, and data visualization. The details of SNA are described in the Supplemental Appendix 6.

We identified themes for each word cluster by combining two methods: analyzing the top words identified by SNA and examining sample posts within each cluster for contextual insights. This process was manually conducted by two authors through in-depth reading and discussion. For instance, the theme for TP_C4 "privacy concern about voice assistants in various contexts" was determined through this process. Initially, key terms in the cluster, such as "assistant," 'voice "intelligent," and "digital," hinted at discussions surrounding voice assistants. Further scrutiny of sample posts revealed these discussions spanned diverse AI applications, including education, healthcare, in-car systems, and customer service. While certain deployment contexts were not immediately apparent from the top words, they became clearer upon deeper examination of the associated posts.

Qualitative Discourse Analysis

To derive a contextual understanding of the word clusters generated by SNA, we further conducted discourse analysis on sampled social media posts. Discourse analysis can reveal the power dynamics, ideologies, and assumptions that underlie language use, providing insight into societal structures and power relationships (Fairclough, 2003). We conducted the analysis on a random sample of 100 posts from each word cluster, totaling 889 tweets and 900 Weibo posts. For a detailed account of our sampling methods, see the Supplemental Appendix 7. Two authors engaged in a meticulous review of these sampled posts, with attention to aspects such as word choice, sentence structure, sentiment, source of the post, and social context. Our analysis sought to uncover the deeper layers of meanings of the sampled posts, grounding them in the sociocultural, economic, and political contexts in the United States and China.

Results

Descriptive Analysis

Out of the 1,165 privacy-related tweets, the majority (81.3%) are contributed by public users, whereas a smaller proportion (18.7%) are attributed to institutional users. The Weibo corpus presents a slightly different picture, with 56.3% posts published by public users, and the remaining (43.7%) posts created by institutional users (see Table 2).

Four semantic networks are constructed based on public and institutional discourses on Weibo and Twitter, respectively, which showcase the structure of the network and major clusters (themes) within these discourses (Figures 1 and 2). In conducting the SNA, we excluded words "privacy" and "隐私" ("privacy" in Chinese) to spotlight words that co-occurred with privacy. We calculated and visualized the centrality of the words within each network so as to highlight the important words in the discourse and uniqueness of each cluster (Segev, 2021). The top 30 central words in all four networks can be found in Supplemental Appendix 8. We summarized the theme of each cluster based on the top words and sample posts and the percentage of each cluster in the whole network (see Table 3).

RQ1: Public Discourse of AI Privacy in the United States and China

Semantic Network Analysis. Both Twitter and Weibo semantic networks feature five themes/clusters, respectively. On Twitter, as shown in Table 3 and Figure 1a, the largest cluster **TP_C1**, accounting for 38.33% of the network, investigates public concerns stemming from Alexa's privacy scandal, featuring keywords like "alexa," "listen," "home," and "concern." The second largest cluster, **TP_C2** (33.17%), encapsulates words indicative of users' privacy concerns over voice assistants and their parent tech companies, with keywords like "siri," "apple," "google," and "surveillance." The third cluster, **TP_C3** (17.44%), is characterized by words associated with cybersecurity and privacy protection, "cybersecurity," "gdpr," "scammer," and "databreach." **TP_C4** (7.13%) addresses privacy concerns about voice assistants' deployment in various contexts, particularly education and healthcare. **TP_C5** delves into concerns and fears incited by Microsoft' virtual assistant, Cortana, denoted by words like "cortana," "fear," "microsoft," and "nadella."

On Weibo (see Figure 1b), the largest cluster, **WP_C1** (33.55% of network contributions), revolves around users' privacy concerns and experiences with conversational AI, with keywords like "user," "information," "voice," "product," and "function." **WP_C2** (28.39%) zooms in on the industrial implications of conversational AI, evidenced by words such as "technology," "corporate," "development," and "enterprise." **WP_C3** (15.7%) is centered on the societal implications of AI and robotics, denoted by words like "artificial intelligence," "robot," "algorithm," "life," and "work." **WP_C4** (15.49%) emphasizes the application scenarios of conversational AI, underscored by words like "service," "device," "platform," "scenario," and "experience." The smallest cluster, **WP_C5** (6.88%) pertains to data protection and regulation, characterized by words such as "data," "collect," "analysis," "regulation," and "third-party."

Table 2. Descriptive Statistics of Twitter and Weibo Data by User Type.

User type	Twitter		Weibo	
	Count	Percent	Count	Percent
The public	947	81.29	638	56.26
Institutions	218	18.71	496	43.74
Corporation	121	10.39	280	24.69
Media	84	7.21	130	11.46
Government	1	0.09	51	4.50
NGO	5	0.43	28	2.47
School	7	0.6	7	0.62
In total	1,165	100	1,134	100

Discourse Analysis. Public discourse on conversational AI privacy on both platforms is rife with concerns about excessive data access, use, and sharing. However, notable differences were evident in the focus areas of Chinese and U.S. users. U.S. users' discussions predominantly revolve around the perceived privacy threats posed by voice assistants such as Amazon Alex. This recurrent theme is also evident in the four thematic clusters in Twitter semantic network (TP_C1, TP_C2, TP_C4, TP_C5), collectively representing over 80% of the network, which are all related to public reactions and privacy concerns regarding the use of voice assistants across various scenarios. Their concerns and frustrations stem from a perceived infringement on individual rights:

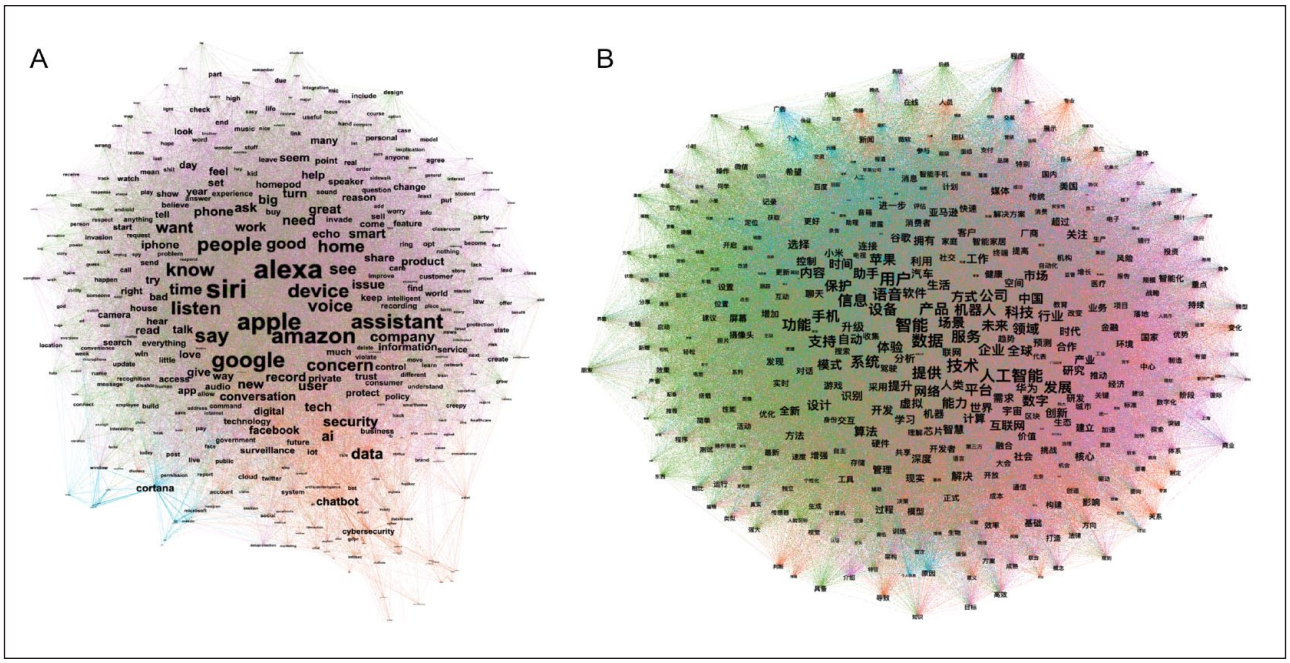

Figure 1. The semantic network visualization of public discourses about conversational AI privacy on (A) Twitter (above) and (B) Weibo (below).
Note. Different colors represent words in different clusters. The size of the word refers to its degree of centrality in the whole network.

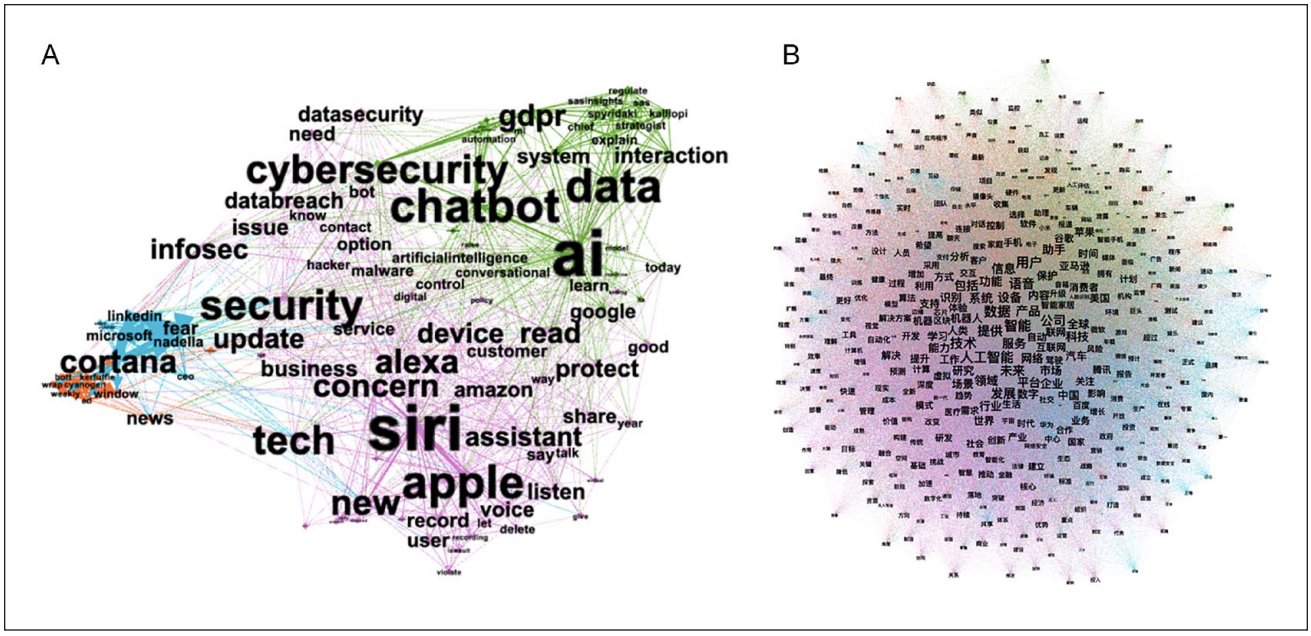

Figure 2. The semantic network visualization of institutional discourses about conversational AI privacy on (A) Twitter (above) and (B) Weibo (below).

P1: "*A new lawsuit accuses Apple of violating user's privacy by allegedly allowing Siri to record without consent . . . DATA PRIVACY RIGHTS! We need laws that protect private DATA, Thank you!*"

P2: "*So my apprehension on Alexa and all the others was that the government, hackers, thieves, etc. could potentially infiltrate your personal home privacy via audio. Now the new ones have cameras and video screens!!! The*

Table 3. Themes of Public Discourse on Conversational AI Privacy on Twitter and Weibo.

Network	Cluster	Theme	Top words (based on degree centrality[a])	Share of the network (%)
Twitter public	TP_C1	Privacy concerns about Amazon Alexa	alexa, amazon, people, device, listen, home, concern, company, phone, record, issue, turn, conversation, echo, talk, product, share, help, set, information, hear, private, protect	38.33%
	TP_C2	Privacy concerns about voice assistants	siri, apple, google, user, smart, work, iphone, app, homepod, audio, surveillance, speaker, point, search, service, feature, policy, win, recording, experience	33.17%
	TP_C3	Cybersecurity and privacy protection	chatbot, cybersecurity, gdpr, email, scammer, scam, databreach, ai, inf, facebook, data, defstar5, tech, iot, makeyourownlane, mpgvip, love, infose, control, infosec, security, technology, bot, github, javascript, conversational, twitter, system	17.44%
	TP_C4	Privacy concerns about voice assistants in various contexts	assistant, voice, digital, intelligent, command, kid, problem, facial, recognition, classroom, student, learn, episode, tool, voicefirst, skill, useful, scary, rule, grow, threat, class, connect, power, creepy, enough, industry, independent	7.13%
	TP_C5	Privacy concerns about Cortana	cortana, fear, linkedin, microsoft, nadella, window, cyanogen, weekly, wrap, bott, kerfuffle, ceo, satya, face, busy	3.39%
Weibo public	WP_C1	Privacy concerns and user experiences with conversational AI	user (用户), information (信息), voice (语音), product (产品), system (系统), function (功能), support (支持), mobile phone (手机), protection (保护), apple (苹果), assistant (助手), content (内容), mode (模式), design (设计), automatic (自动), time (时间), software (软件), upgrade (升级), choice (选择), own (拥有), google (谷歌), brand new (全新), control (控制), amazon (亚马逊), xiaomi (小米), game (游戏), screen (屏幕), increase (增加), drive (驾驶), consumer (消费者)	33.55%
	WP_C2	Industrial implications of conversational AI	technology (技术), company (公司), science and technology (科技), development (发展), field (领域), enterprise (企业), future (未来), network (网络), digital (数字), industry (行业), China (中国), improve (提升), market (市场), Internet (互联网), virtual (虚拟), computation (计算), research (研究), innovation (创新), attention (关注), era (时代), trade (产业), universe (宇宙), space (空间), core (核心), networking (联网), adopt (采用), business (业务), research and development (研发), United States (美国), value (价值), exceed (超过)	28.39%
	WP_C3	Societal implications of AI and robotics	artificial intelligence (人工智能), robot (机器人), approach (方式), algorithm (算法), life (生活), human (人类), work (工作), utilize (利用), recognize (识别), solve (解决), learn (学习), machine (机器), deep (深度), method (方法), society (社会), chat (聊天), media (媒体), health (健康), model (模型), conversation (对话), improve (提高), socialize (社交), tool (工具), formal (正式), center (中心), medical (医疗), news (新闻)	15.7%
	WP_C4	Application scenarios of conversational AI	intelligent (智能), provide (提供), service (服务), device (设备), platform (平台), context (场景), experience (体验), capability (能力), global (全球), world (世界), huawei (华为), automobile (汽车), development (开发), demand (需求), smart (智慧), cooperation (合作), chip (芯片), connect (连接), reality (现实), prediction (预测), hardware (硬件), interaction (交互), manufacturer (厂商), ecosystem (生态), developer (开发者), enhance (增强), smart home (智能家居), construct (构建), family (家庭)	15.49%
	WP_C5	Data protection and regulation	data (数据), collect (收集), analyze (分析), understand (理解), promote (推动), build (建设), third party (第三方), discover (发现), client (客户), regulation (监管), principle (原则), challenge (挑战), institution (机构), decision-making (决策), communicate (沟通), define (定义), criteria (标准), data analysis (数据分析), precise (精准), establish (建立), real-time (实时), unified (统一), trust (信任), sales (销售), management (管理), personnel (人员), accelerate (加快), process (过程), speed (速度), personal information (个人信息), reason (原因), marketing (营销)	6.88%

[a]Degree centrality is calculated based on the number of links a word has with other words, which indicates the popularity of a word. A word with a high degree of centrality means it is frequently mentioned with many other words.

bad guys are about to see it all! Big brother is watching! #1984"

Conversely, Chinese users often talk about AI privacy in a broader context, focusing on its benefits and risks to the entire society rather than just individual users. Their discourse highlights the societal implications of privacy issues with emerging technologies, reflecting China's more collectivist culture. This is evidenced by three thematic clusters (WP_C2, WP_C3, WP_C4), collectively representing nearly 60% of the network. These clusters feature words that represent the broader industrial and societal implications of privacy issues in AI and are characterized by high-level, abstract, and ambiguous language. For example, P3 underscores potential privacy risks across various digital contexts, emphasizing the implications not just for individuals but also for the collective online community.

P3: "*#Home Camera Suddenly Speaking# For example, if you install a camera on top of Xiaoai's head, Xiaoai will activate by itself . . . As the features of cars are gradually extended, eavesdropping, privacy, information, safe driving, and a series of issues will profoundly affect the ethics and laws of our society. Netizens are so innocent.*"

Despite a Chinese thematic cluster (WP_C1, 33.55%) devoted to individual-level experiences with conversational AI, Weibo users often discuss these matters in an ambiguous manner, with some claiming that they do not have privacy as an ordinary person (see P4 below). This reflects a general lack of understanding and awareness of privacy in Chinese society.

P4: "*I have never encountered a situation where my data was leaked or the system crashed. As for whether Siri is listening to my privacy, I'm not sure (it seems like I don't have much privacy to begin with, just an ordinary person)*."

Cultural nuances are also evident in the sentiments and tones of user expressions. U.S. users display a more emotional and confrontational tone in conveying their fears and anxieties, whereas Chinese users' expressions were more neutral and contemplative. These observations can be explained by Hofstede's cultural dimensions of individualism and collectivism pertaining to communication style. Individualist societies like the U.S. value direct communication and may perceive confrontation as constructive, whereas collectivist societies such as China uphold harmony, favoring a more indirect mode of articulating dissent or concern (Hofstede, 1984). In addition, the less critical discourse observed on Chinese social media could be attributed to online censorship and state surveillance, which may limit the scope and tone of discussions (King et al., 2013). This environment can result in a more restrained and cautious public discourse than the more open and critical expressions seen in the United States.

Although both Chinese and U.S. users share real-life experiences of privacy invasions by conversational AI systems, such as Alexa and Xiaoai, their responses to these experiences differ. U.S. users recount these experiences and link them to specific actions or behavioral changes, often advocating for privacy protection laws and regulations and collective resistance against privacy infringements (e.g., P5). A significant Twitter word cluster TP_C3 (comprising 17.44% of the network) elucidates this proactive stance, indicating a rich discourse on practical measures to bolster privacy and cybersecurity. In addition, as seen in P6, Twitter users utilize privacy-related hashtags to engage in and stir public discourse, reflecting the role of individual advocacy in raising public awareness about privacy issues.

P5: "*I HATE CORTANA . . . WHY HASN'T THERE BEEN A CLASS ACTION LAWSUIT AGAINST THE COLLECTION OF PRIVACY DATA?*"

P6: "*Been testing this out the last couple of days, and it's a bit disturbing how often a random topic out in brief convo shows up as a targeted ad the next day. Siri is always listening #bigbrother #iphonespies #theyarelistening #privacymatters*"

Conversely, Chinese users seldom share actions taken to protect their privacy, despite descriptions of a discomfort with the unpredictable behavior of voice assistants and potential surveillance (see P7). This implies a tendency toward raising awareness rather than initiating action within the Chinese context. The smaller WP_C5 cluster (7%) also substantiates a lesser focus on privacy protection behavior among Chinese users.

P7: "*Today we're still discussing the privacy issues of voice assistants at work. Are they listening to us 24/7 through various voice triggers/awakening methods? This morning I just discussed The Three-Body Problem with a colleague, and as soon as I opened Zhihu, it pushed a topic on The Three-Body Problem? It's a bit terrifying when you think about it*."

It is important to note that both Twitter and Weibo include individual users who are professional opinion leaders, potentially merging public and institutional discourses. While Twitter did not categorize different types of users, Weibo provided a built-in classification of ordinary users and opinion leaders (big-V users). Thus, we conducted a post hoc analysis on semantic networks of discourse from ordinary users (60% of the total posts) and big-V users (40%) on Weibo. Our results revealed a notable thematic consistency between the discourses of big-V users and ordinary users, with a focus on industrial and societal implications of conversational AI, advancements in China's AI sector, and reposts of news coverage of data breaches involving major

Table 4. Themes of Institutional Discourse on Conversation AI Privacy on Twitter and Weibo.

Network	Cluster	Theme	Top words	Share of the network (%)
Twitter Institution	TI_C1	Corporate privacy scandals	siri, apple, security, tech, alexa, concern, device, assistant, protect, business, databreach, amazon, datasecurity, voice, record, share, user, customer, option	44.66%
	TI_C2	Privacy-preserving tools and strategies	Ai, artificialintelligence, automation, bot, chatbot, conversational, cybersecurity, email, google, hacker, infosec, learn, machine, malware, ml, model, policy, scam, scammer, service, setting	28.16%
	TI_C3	Data regulation and protection	Chief, data, explain, gdpr, interaction, kalliopi, regulate, sas, insights, spyridaki, strategist, system	11.65%
	TI_C4	Privacy concerns about Cortana	Cortana, fear, linkedin, microsoft, nadella, ceo, orlando, satya	7.77%
	TI_C5	System/software updates	News, window, bot, cyanogen, kerfuffle, weekly, wrap	7.77%
Weibo Institution	WI_C1	AI advancement and application	data (数据), artificial intelligence (人工智能), technology (技术), provide (提供), development (发展), future (未来), field (领域), enterprise (企业), platform (平台), industry (行业), network (网络), ability (能力), global (全球), China (中国), scenario (场景), research (研究), recognition (识别), number (数字), learning (学习), human (人类), world (世界), method (方式), work (工作)	52.32%
	WI_C2	Smart devices and IoT	intelligence (智能), device (设备), speaker (音箱), function (功能), system (系统), automatic (自动), driving (驾驶), service (服务), interaction (交互), product (产品), internet (联网), market (市场), automobile (汽车), experience (体验), smart home (智能家居), chip (芯片), edge (边缘), home (家庭), life (生活), Baidu (百度), television (电视)	23.40%
	WI_C3	U.S. corporate privacy scandals	assistant (助手), voice (语音), user (用户), apple (苹果), recording (录音), amazon (亚马逊), google (谷歌), report (报道), information (信息), protection (保护), content (内容), listen (听取), company (公司), choice (选择), instruction (指令), website (网站), upgrade (升级), leak (泄露), technology (科技), assistant (助理), collect (收集), consumer (消费者), USA (美国), prevent (阻止), track (跟踪), infringe (侵犯), exhibition (展会), Cook (库克)	21.63%
	WI_C4	Privacy concerns about robots and chatbots	robot (机器人), internet (互联网), increase (增加), dialogue (对话), chat (聊天), team (团队), search (搜索), open (开放), new generation (新一代), Musk (马斯克), long term (长期), comment (评论)	2.65%

tech companies. Detailed results are available in Supplemental Appendix 9.

RQ2: Institutional Discourse About Conversational AI privacy in the United States and China

Semantic Network Analysis. Table 4 delineates the semantic clusters derived from Twitter and Weibo institutional discourse. On Twitter (see Figure 2a), the dominant cluster, **TI_C1**, accounting for 44.66% of the semantic network, concerns data security and privacy issues surrounding voice assistants, with keywords such as "siri," "alexa," "security," "databreach," and "protect." **TI_C2**, the second largest cluster (28.16%), investigates the application of AI tools and chatbots in identifying and mitigating cybersecurity threats, with terms such as "cybersecurity," "malware," and "scams." **TI_C3** (11.65%) focuses on data regulation and protection, specifically in relation to the GDPR. The discourse revolves around the news of Kalliopi Spyridaki, Chief Strategist at SAS, offering insights into GDPR's influence on system regulations. **TI_C4** (7.77%) centers on privacy concerns about Microsoft's Cortana, and **TI_C5** (7.77%) highlights news of system/software updates.

Figure 2b displays the four thematic clusters of Weibo institutional discourse. The largest cluster **WI_C1** (52.32%) examines global and Chinese development and application of AI and related technologies, including keywords like "artificial intelligence," "development," "global," "China," "enterprise," and "research." **WI_C2** (23.40%) centers on smart devices, smart home technology, and the Internet of

Things (IoT), exploring their functionality, user experience, and user-device interaction. Seminal words include "device," "driving," "interaction," "experience," and "smart home." **WI_C3** (21.63%) scrutinizes data breaches and privacy violation events regarding voice assistants from major U.S. tech companies, featuring words such as "recording," "violation," "leakage," "Apple," "Google," "Amazon," and "U.S." **WI_C4** (2.65%) explores privacy concerns over robots and chatbots, driven by a tweet about an AI chatbot with long-term memory and internet searching capabilities developed by Facebook's AI team.

Discourse Analysis. A common theme shared across both platforms concerns privacy violations by U.S. tech giants, such as Apple, Google, and Amazon. This theme accounts for over 50% of the discourse in two Twitter clusters (TI_C1, TI_C4) and 25% in two Weibo clusters (WI_C3 & WI_C4). While the institutional discourse around AI privacy on Weibo and Twitter shares similar themes, the presentation and tone of these discussions exhibit stark contrasts.

U.S. institutional Twitter accounts are notably more critical of these incidents. They urge tech companies to assertively protect user privacy and encourage users to proactively secure their data. P1, for example, references a Business Insider article about a significant Alexa data breach. This tweet employs a range of hashtags to situate the incident within broader digital privacy and data security discourses, subtly critiquing tech companies' security practices with the claim that "Smart Devices keep data!."

> P1: "*Amazon accidentally sent 1700 Alexa recordings to the wrong person, including audio from in the shower |@ businessinsider Smart Devices keep data! CyberSecurity #Security #DataPrivacy #DataBreach #Technology #Business #Ransomware #Malware #Tech #Data #NYC*"

In contrast, Chinese institutional accounts on Weibo adopt a more neutral tone, offering factual reports on privacy violation incidents. However, considering the unavailability of Facebook, Amazon, and Google in China and the existing Sino-U.S. tensions, these posts subtly underscore the risks associated with foreign companies. This potentially plays into a larger narrative of caution or skepticism toward U.S. tech giants, which could be influenced by the prevailing geopolitical tensions. A prevalent example includes a viral post from the Chinese Central Television's (CCTV) financial channel, the government's official broadcaster. The post, primarily an information report, reveals investigations into Google, Apple, and Amazon by European and U.S. regulatory bodies for alleged privacy violations.

> P3: "*According to a report on the Fortune website on the 5th, regulatory and legislative bodies in Europe and the United States are investigating Google, Apple, and Amazon, scrutinizing whether these three companies have allegedly violated user privacy . . . Amazon, under the pretext of improving software quality, has been listening to all command recordings issued by users to the voice assistant Alexa . . .*"

By highlighting the privacy violations of Alexa and Google Voice Assistant, which are unavailable in China, the post could be seen as reinforcing the Chinese government's restrictions on these services. This might indirectly validate the Chinese government's stance of protecting its citizens' privacy and national security from foreign technologies. While the post appears to be a simple sharing of news, it likely reflects and contributes to larger dialogues surrounding technology, privacy, and geopolitical tensions.

Contrasting attitudes toward AI technologies and privacy issues emerge from the institutional discourse in the United States and China. U.S. institutions often express caution and concern about AI and privacy, aiming to enhance public awareness and literacy on the matter. They provide practical strategies for users to secure privacy in the use of conversational AI, evident in Twitter's second-largest cluster (TI_C2). Strategies vary from tweaking privacy settings, halting data sharing with third parties, to employing technical solutions like federated learning. Apple Support, for instance, contributes to this discourse, advising users on ways to boost privacy, including disabling Siri options and deleting Siri history.

Hashtag activism was also observed on Twitter, with hashtags like #DataPrivacyDay, #privacyprotections, and #techwellness enhancing visibility of privacy and cybersecurity issues and augmenting public privacy literacy. @FirestormCyber, a security services provider, exemplifies such proactive communication:

> P4: "*Upgrading your iPhone or iPad to 13.2? Protect your privacy! Read the fine print and Just Say No unless you want Apple to listen to any voice interactions you have with Siri. #JustSayNo #privacymatters #privacy #YourDeviceIsListening #Apple #iOS*"

Institutions also promote platforms and tools with strong privacy protection measures, presenting privacy protection as a unique selling point. P6 showcases a chatbot platform prioritizing privacy protection and employs relevant hashtags to boost visibility.

> P5: "*The whole Engati chatbot platform is now available on-premise. No privacy issues, no compliance issues, no third parties. Gain complete control & build as many bots as you need, with Engati On-premise. #datasecurity #onpremiseChatbots #automation #ml*"

Conversely, in China, institutional discourse leans toward acceptance and usage of AI technologies, with less focus on individual privacy concerns. This is reflected in the two predominant word clusters on Weibo (WI_C1, WI_C2), accounting for over 75% of the discourse, that underscore advancements and applications in AI. Numerous posts delve

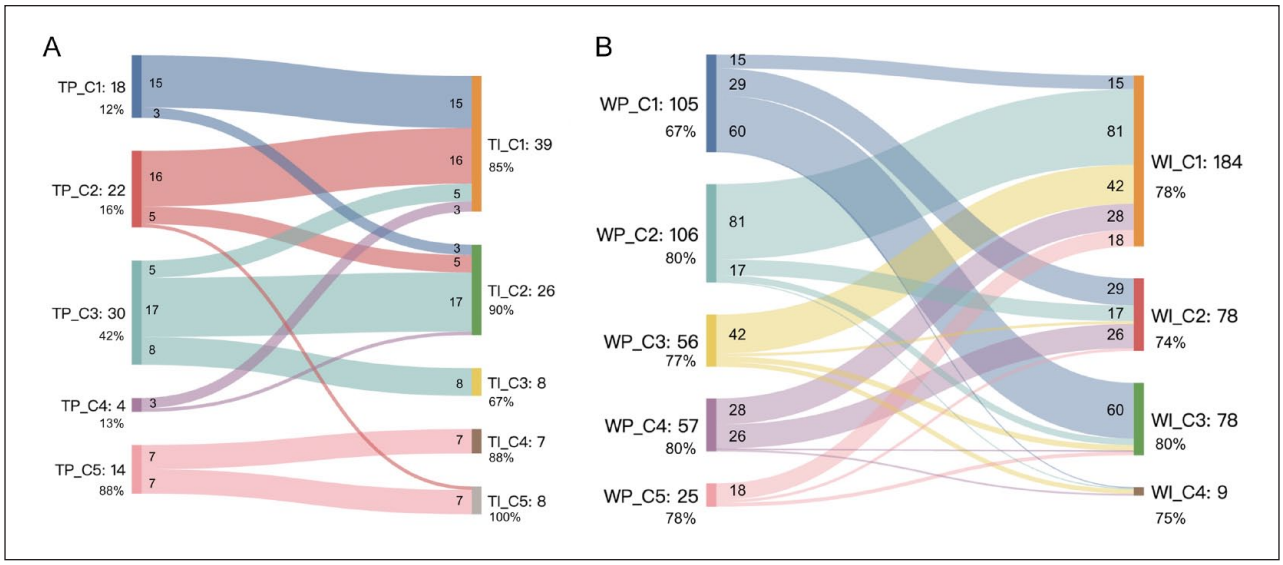

Figure 3. Sankey flow diagram of shared words between public and institutional word clusters on (A) Twitter and (B) Weibo [1,2].

into the transformative role of AI across various sectors and their ethical implications. Notably, discussions on ethical issues of AI, including data privacy and security, are often introduced toward the end. This sequence perhaps mirrors a deeper narrative within China's expansive tech landscape and national AI initiative—a narrative that traditionally champions swift technological progress, placing ethical deliberations as subsequent considerations. This is exemplified in P6:

> P6: "*Huawei has released its 'Top Ten Trends Towards 2025.' Trend Ten: Huawei calls for a global acceleration in establishing unified data standards and principles of data use. It also encourages the promotion of third-party data regulatory bodies to ensure compliance with privacy, security, and ethics, providing a legal basis for these principles.*"

In summary, the institutional discourses on Twitter and Weibo reveal divergent emphases, meanings, and ideologies, which are likely attributed to different social, cultural, and political contexts in the United States and China.

RQ3: Convergence and Divergence of Public and Institutional Discourse

To elucidate the convergence and divergence between public and institutional discourse on the two platforms pertaining to the U.S. and Chinese contexts, Sankey flow diagram (Figure 3) was deployed to showcase the number of shared words across the word clusters identified within public and institutional corpus.

Twitter's discourse appears more diverse, with most public clusters sharing only 12%–42% of words with institutional clusters, indicating a wide spectrum of perspectives. Twitter reveals a notable intersection between public cluster TP_C3 and institutional cluster TI_C2, suggesting a mutual focus on cybersecurity in relation to AI chatbots, underscoring the importance both groups attribute to privacy protection. The public clusters TP_C2 and TP_C1, which emphasize privacy risks of conversational AI, intersect with the institutional cluster TI_C1, creating an interactive space for public sentiment and institutional narratives. Such overlaps suggest a mutual influence between public expressions of concern about voice assistants and institutional discourse on broader privacy issues.

Another interesting finding from our analysis reveals that while public Twitter users advocate for system solutions like industrial regulation and legislation, institutional Twitter discourses appear to propel a more individualistic approach toward privacy protection. This emphasis on "self-help" could be seen as reflective of a broader cultural value that highlights personal responsibility and autonomy. While this stance may empower users by encouraging individual agency, it may risk deflecting accountability away from institutions, which could potentially downplay the necessity of wider systemic changes in privacy practices and regulations.

Conversely, on Weibo, a considerable overlap occurs within word clusters, indicative of a high convergence between public and institutional narratives. The shared words comprise between 67% and 80% of the public discourses and between 74% and 80% of the institutional discourses. Specifically, the public clusters WP_C2 and WP_C3, which discuss privacy implications of AI, align notably with the institutional cluster WI_C1, suggesting institutional narratives on AI's transformative potential are shaping public recognition of AI's privacy implications. Similarly, a substantial overlap exists between public cluster WP_C1 and institutional cluster WI_C3, indicating heightened privacy awareness among Chinese users in the wake of high-profile privacy scandals involving U.S. tech giants.

The contrast between Twitter and Weibo reflects the cultural and political differences in these two countries. In the United States, an individualistic norm may foster diverse viewpoints and increased skepticism toward institutional narratives, whereas Weibo seems to promote greater narrative harmony, possibly due to a collectivist cultural norm and authoritarian societal facets. The overlap in discourse can also be viewed through the lens of China's political environment. The Chinese government exerts significant control over media and public communication, leading to a high degree of alignment between public and institutional narratives. The convergence observed in the word clusters suggests that institutional messages about AI, likely disseminated through state-controlled media, are effectively shaping public perceptions. This indicates the success of the government in steering public discourse to align with state policies and objectives regarding AI and privacy.

Discussion

Through a mixed-methods analysis of social media discussions on privacy issues regarding conversational AI, this study revealed notable differences in public and institutional discourses on Twitter and Weibo, highlighting varying degrees of convergence between public and institutional narratives. Such differences are intertwined with the cultural, political, economic, and regulatory contexts in the United States and China. These insights can guide more effective communication strategies, regulatory approaches, and public engagement efforts. We discuss the main findings and contributions of this study below.

Contextualizing AI Privacy Beyond Nation and Culture

Comparative research seeks to identify general patterns and validate them through contextualization, avoiding normalization and universalization (Esser & Hanitzsch, 2012). Yet prior comparative communication research has mainly viewed country or nation as the unit of comparison and often relied on Hofstede's cultural dimensions as interpretative frameworks (Cho et al., 2009; Dogruel & Joeckel, 2019). Echoing the CPR framework, we believe that focusing on more specific units of comparison, such as cultural values, social norms, political and economic systems, and legal/regulatory frameworks, will provide a deeper understanding of the complex dynamics influencing privacy.

Our analysis unveiled divergent meanings, perceptions, and practices regarding conversational AI privacy in the United States and China. In the United States, privacy is often viewed through the lens of individual rights, whereas in China, privacy is more closely tied to collective welfare and state security. Moreover, public discourse on Twitter was marked by direct experiences of privacy risks and proactive privacy-protective actions, whereas Weibo discourse often revolved around indirect privacy risks and broader discussions of AI's societal implications. While cultural values may play a role (e.g., Cho et al., 2009; Trepte et al., 2017), such differences also likely stem from the varied sociopolitical, economic, and legal contexts in these countries. In the United States, the emphasis on individual rights, democratic debate, and market competition fosters a critical and diverse public discourse on privacy (Cohen, 2000; Gavison, 1980; Solove, 2002). In contrast, China's collectivist values, authoritarian political system, and state-driven economic strategies promote a more aligned and supportive narrative around AI privacy, reflecting broader societal goals and national priorities (King et al., 2013). Our findings also highlight the potential impact of regulatory differences, with U.S. discussions often advocating for comprehensive privacy laws, whereas Chinese discourse supports state-led initiatives.

This study also revealed variations in institutional discourse on privacy across countries, which is often overlooked in previous research (e.g., Liang et al., 2017; Ma, 2023). Institutional discourse on Twitter frequently critiques the privacy risks associated with conversational AI and major tech companies, highlighting a proactive approach to addressing privacy concerns, often driven by the need to respond to public scrutiny and regulatory pressures. In contrast, institutional discourse on Weibo tends to underscore the technological advancements and acceptance of AI in China, reflecting a more positive and supportive narrative around AI, aligning with the government's strategic goals to position China as a leader in AI technology (Wu et al., 2020). This finding contributes to the broader understanding of how cultural, political, and market dynamics may shape institutional privacy practices in different contexts.

This research is one of the few studies that examine comparative privacy across multiple units of comparison, illustrating how privacy perceptions are entwined with various contextual factors. Our findings also show that these factors are interconnected, presumably shaping the privacy discourse in complex ways. We emphasize the need to transcend national boundaries and look beyond national cultures in CPR, embracing the complexities and dynamic nature of privacy in the digital age. This approach can help generate more nuanced and durable insights into privacy perceptions, norms, and practices.

Moving Toward Multi-Level CPR

McLeod and Lee (2013) argue that comparative communication research often relies on single-level analysis, attributing differences to cross-national variations, resulting in incomplete and potentially misleading explanatory models. This limitation is also evident in CPR, as noted by Masur et al. (2021). Our study fills this gap by examining both public and institutional discourses and exploring their convergence and divergence. This multi-level approach reveals the complex

interplay between different levels of discourse and how they collectively shape privacy norms and practices.

Our findings shed light on the interplay between individual discourse (micro-level), institutional discourse (meso-level), and broader contexts (macro-level). A low overlap between public and institutional discourses on Twitter indicates a diverse and fragmented privacy discourse, reflecting a democratic, individualistic culture in the United States that values open debate and advocacy for systemic change. In contrast, the high convergence between public and institutional discourse in China reflects a collectivist culture and an authoritarian political environment that promotes alignment with institutional narratives and social harmony. Previous studies have shown that government and corporate surveillance may result in digital resignation and privacy apathy (Draper & Turow, 2019; Zuboff, 2019). Our findings suggest that institutional influence on digital resignation appears to be more pronounced in China than in the United States. Micro-level privacy concerns and practices in the United States are more independently expressed and debated, while in China, they are closely intertwined with meso- and macro-level factors.

Therefore, we suggest that a multi-level analysis in CPR is invaluable. Macro-level factors encompass cultural, political, economic, and regulatory contexts that shape the overall environment for privacy practices. Meso-level factors refer to organizational practices and policies, such as industry standards and corporate privacy policies. Micro-level factors include individual behaviors and perceptions, such as privacy preferences, awareness, and literacy (e.g., Barnes, 2006; Cho et al., 2009, 2023; Dienlin & Trepte, 2015). Each level offers unique insights that, when combined, can offer a comprehensive understanding of privacy. This multi-level approach provides a holistic view of how individual behaviors, organizational practices, and broader regulatory and cultural contexts interact to shape privacy in society. By examining the interplay between factors across levels, we can better understand the complex dynamics at play and develop holistic strategies to address privacy concerns.

Practical Implications

Our findings on the varied privacy discourses in the United States and China have implications for policymakers, tech companies, and individuals. For policymakers, understanding these differences is crucial for developing privacy regulations that are sensitive to public concerns and perceptions. For U.S. policymakers, the public's demand for federal laws suggests the need to develop and enforce comprehensive federal privacy legislation to provide uniform protections and reduce regulatory gaps. Chinese policymakers may need to balance the promotion of technological advancements with the protection of individual privacy rights.

For technology companies, especially those operating in both markets, there is a need to tailor their strategies to local needs and preferences. In regions where privacy is highly valued such as the United States, emphasizing robust privacy protections may enhance competitive advantage and attract consumers. They can invest in privacy-preserving technologies and communicate these efforts transparently to the public. In China, while developing privacy policies that comply with state regulations is crucial, companies should also address user concerns about data security and privacy. Companies can also play a role in educating users about privacy and how to protect their personal information.

For the general public, our study underscores the critical need for engagement with privacy discourse. Post hoc analysis shows a notable lack of public discourse on privacy issues in both countries, with these discussions comprising only 0.15% of all Twitter posts and 0.57% of all Weibo posts related to conversational AI. Engaging critically with privacy discussions enables individuals to navigate the digital world more confidently and securely. The role of hashtag activism in the United States highlights the importance of public engagement and grassroots movements in shaping privacy discourses. Although similar strategies may be less effective in China due to different political and social dynamics, users can still advocate for better privacy practices by providing feedback to companies and participating in dialogues that align with state policies.

Limitations and Future Research

Several limitations of this study should be noted. The main limitation concerns the representativeness and generalizability of the findings. Our focus on a single platform from each country may limit the generalizability of our findings. Future research should collect data from various social media platforms and examine the extent to which privacy-related discussion converge and diverge across platforms within and between countries. It is important to note that, despite the insights social media data provides into AI privacy perceptions, it does not encapsulate the entire spectrum of public discourse on privacy. Online data might be subject to self-selection bias, platform moderation, and government censorship (Lazer et al., 2020). Thus, results from social media data analysis can only reflect a mediated reality rather than genuine human behavior (Wu & Taneja, 2020). Future research could benefit from combining social media data with survey or interview data to triangulate the results and provide a more holistic understanding of AI privacy.

Second, while we employed multiple methods to optimize the retrieval of relevant tweets sent from the United States, we acknowledge that geo-tagging and account-level location may only trace around half of these posts, overlooking tweets without location information. Third, this study implemented a conservative keyword strategy for data relevance and precision, only using the keyword "privacy" to identify privacy-related posts. However, this approach may risk overlooking privacy-related discussions

that did not explicitly use these terms. Future research could employ a more expansive search strategy to capture a wider range of privacy-related discourse.

Last, our study did not delve into the potential variation in privacy discourse across different types of conversational AI systems (e.g., rule-based vs. AI-based, embodied vs. non-embodied, human-like vs. non-human-like systems). This limitation largely stems from our dataset being skewed toward voice assistants. As the landscape of conversational AI diversifies, future research should probe how privacy concerns and practices may converge or diverge across various types of conversational AI systems and other digital technologies (e.g., social media AR, VR).

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Supplemental Material

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Notes

1. In the Sankey diagrams, the numbers within each connecting line (edge) represent the count of common words between two clusters. The numbers on the outer edges represent the number of words a specific cluster shares with clusters on the opposite side. The percentages refer to the proportion of these shared words within the cluster itself. For example, the value 105 besides WP_C1 signified that 105 words are common between WP_C1 and the entire Weibo institutional corpus, which makes up 67% of all words in WP_C1.
2. Abbreviations: **TP** = Twitter public discourse; **TI** = Twitter institutional discourse; **WP** = Weibo public discourse; **WI** = Weibo institutional discourse. **TP_C1** = Public concerns over Amazon Alexa; **TP_C2** = Privacy concerns over voice assistants, **TP_C3** = Cybersecurity and privacy protection, **TP_C4** = Privacy concerns about voice assistants in various contexts, **TP_C5** = Privacy concerns over Cortana, **TI_C1** = Corporate privacy scandals, **TI_C2** = Privacy-preserving tools and strategies, **TI_C3** = Data regulation and protection, **TI_C4** = Privacy concerns about Cortana, **TI_C5** = System/software updates, **WP_C1** = Privacy concerns and experiences with conversational AI, **WP_C2** = Industrial implications of conversational AI, **WP_C3** = Societal implications of AI and robotics, **WP_C4** = Application scenarios of conversational AI, **WP_C5** = Data protection and regulation, **WI_C1** = AI advancement and application, **WI_C2** = Smart devices and IoT, **WI_C3** = US corporate privacy scandals, **WI_C4** = Privacy concerns about FB chatbots.

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